

2016.0 RANGE ROVER (LG), 501-30

REAR END SHEET METAL REPAIRS

INNER QUARTER PANEL UPPER - LONG WHEEL BASE, VEHICLES WITHOUT: GLASS ROOF PANEL (G1676591)

REMOVAL AND INSTALLATION

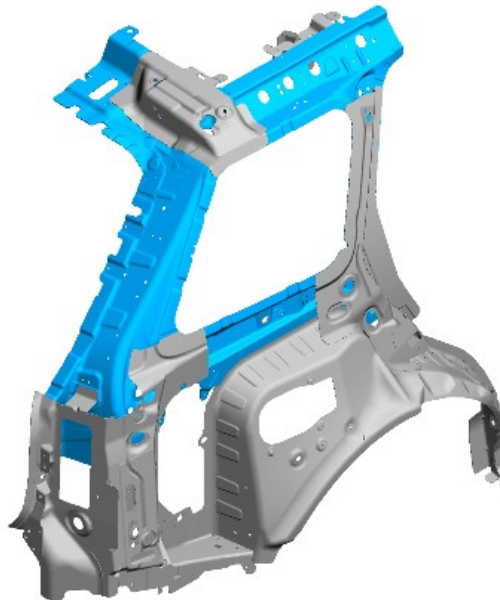
REMOVAL

NOTE:

The inner quarter panel upper is installed in conjunction with:

- Roof rear panel
 - Side panel
 - Quarter panel
 - Rear lamp panel outer
 - Rear lamp panel inner
 - Quarter panel reinforcement
 - Rear wheelhouse outer
 - Rear wheelhouse extension
 - Inner quarter panel lower
 - Water drain panel
 - Roof opening panel
-

1.



E151011

The inner quarter panel upper is serviced as an assembly as indicated.

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2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

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3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

- To make sure that the vehicle is correctly aligned it must be placed on a Land Rover approved alignment jig.

4. For additional information, refer to: [Roof Rear Panel - Standard Wheel Base](#) (501-28 Roof Sheet Metal Repairs, Removal and Installation).
5. For additional information, refer to: [Side Panel - Long Wheel Base](#) (501-29 Side Panel Sheet Metal Repairs, Removal and Installation).
6. For additional information, refer to: [Quarter Panel](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
7. For additional information, refer to: [Rear Lamp Panel Outer](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
8. For additional information, refer to: [Rear Lamp Panel Inner](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
9. For additional information, refer to: [Quarter Panel Reinforcement](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
10. For additional information, refer to: [Rear Wheelhouse Outer](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
11. For additional information, refer to: [Rear Wheelhouse Extension](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
12. For additional information, refer to: [Inner Quarter Panel Lower](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
13. For additional information, refer to: [Water Drain Panel](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).

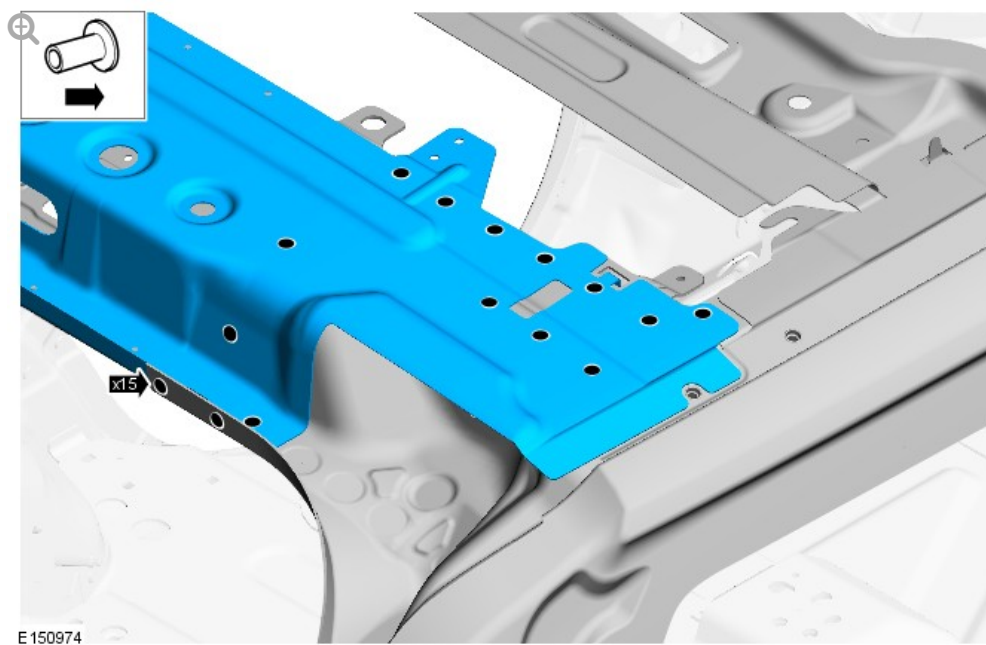
14. For additional information, refer to: [Roof Opening Panel Frame](#) (501-17 Roof Opening Panel, Removal and Installation).

15.



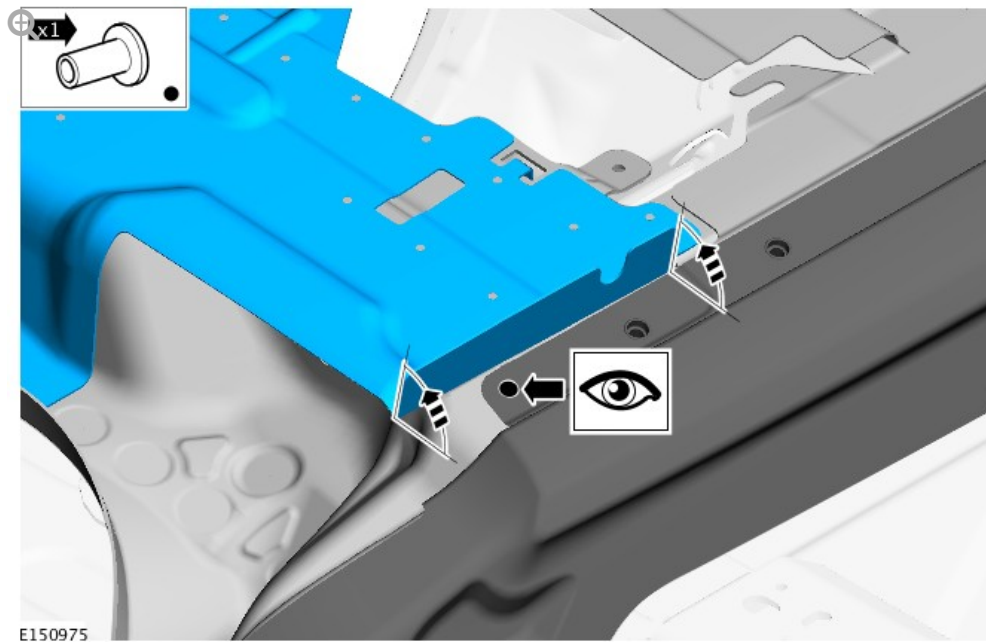
In this procedure the quarter panel requires a cut line on the C-pillar as indicated.

16.



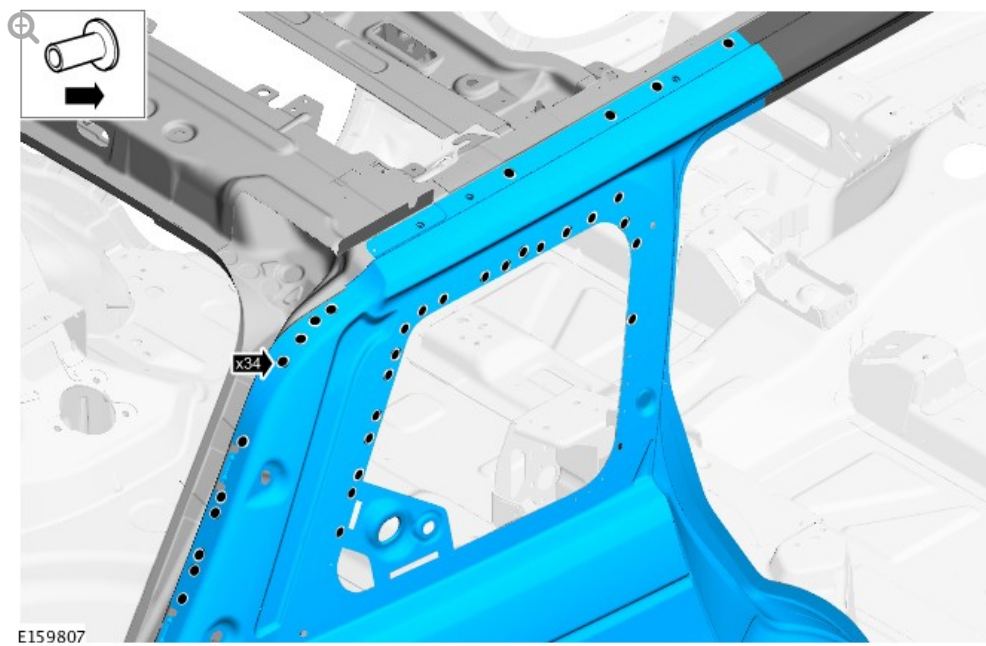
Remove the SPR's as indicated.

17.



Lift the edge of rear header rail to reveal hidden SPR as indicated.

18.



This procedure requires additional SPR's to be removed as indicated.

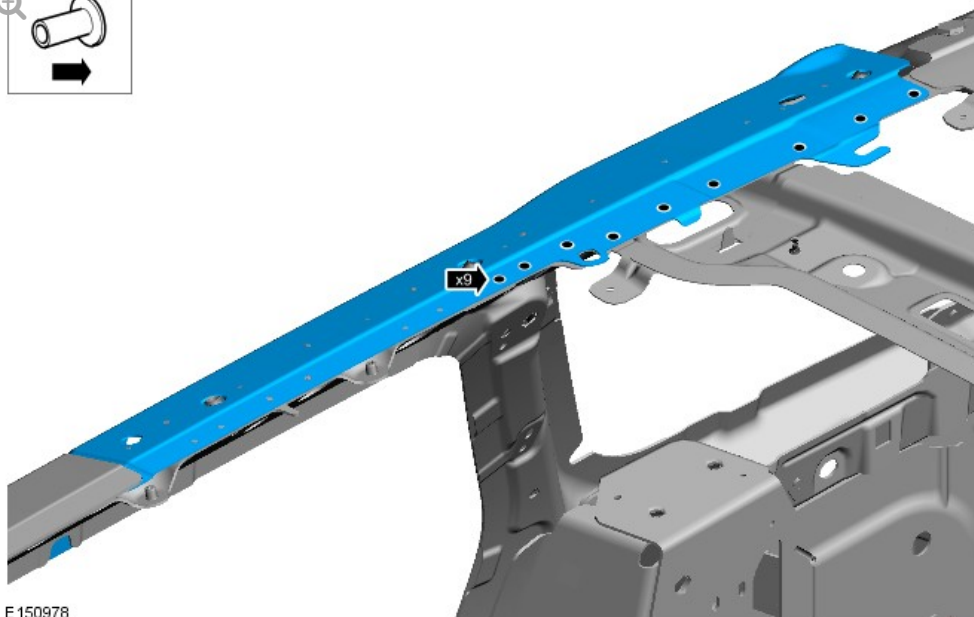
19.



E159808

Remove the SPR's as indicated.

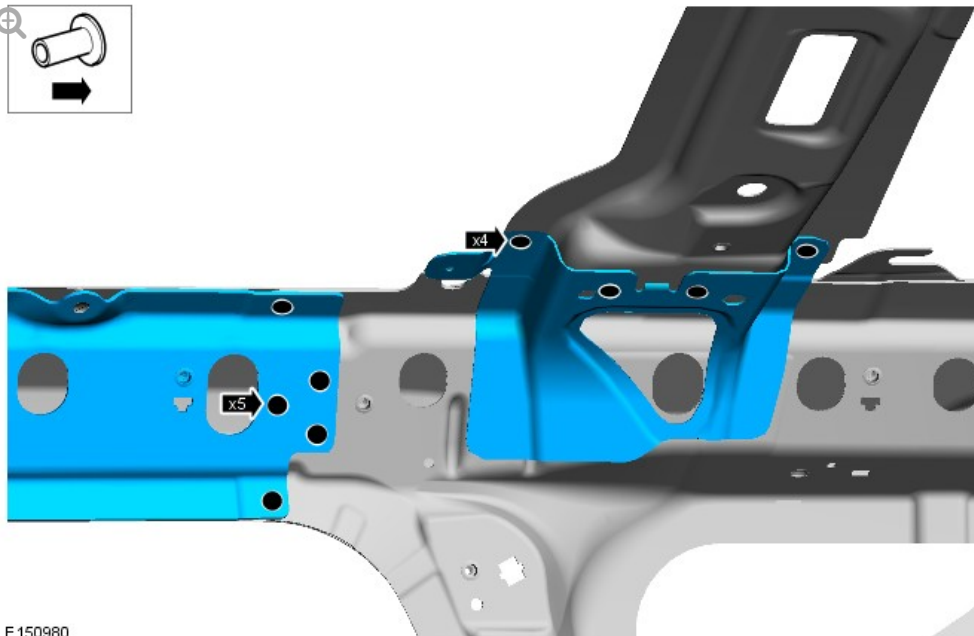
20.



E150978

Remove the SPR's as indicated.

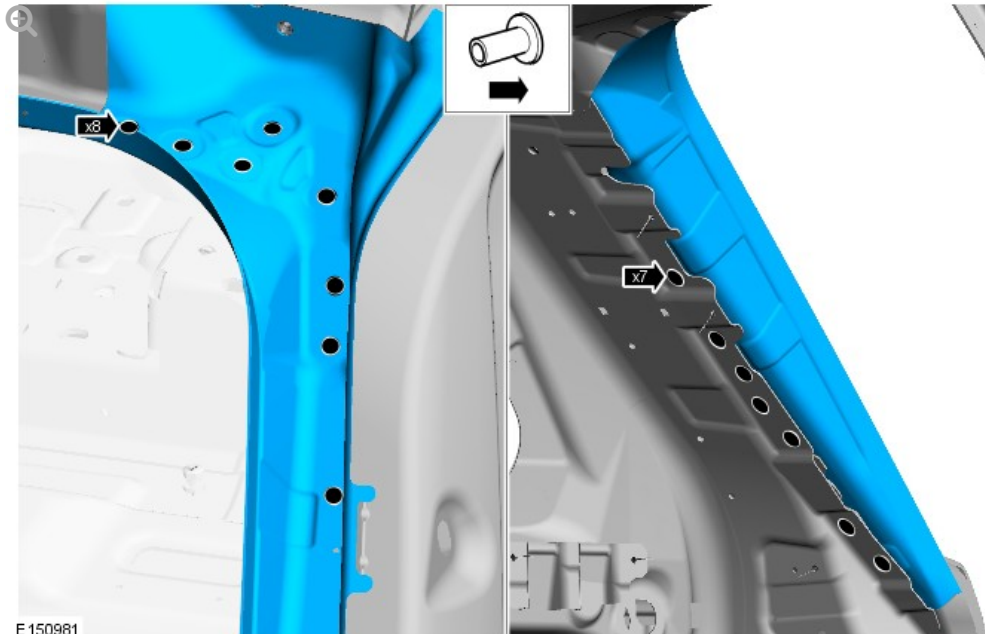
21.



E 150980

Remove the SPR's as indicated.

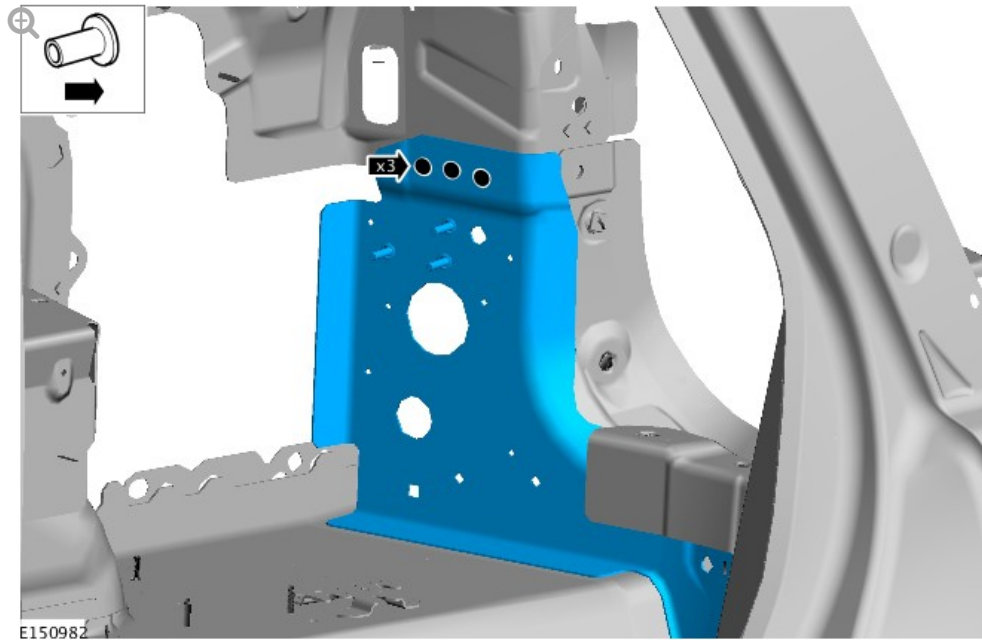
22.



E 150981

Remove the SPR's as indicated.

23.



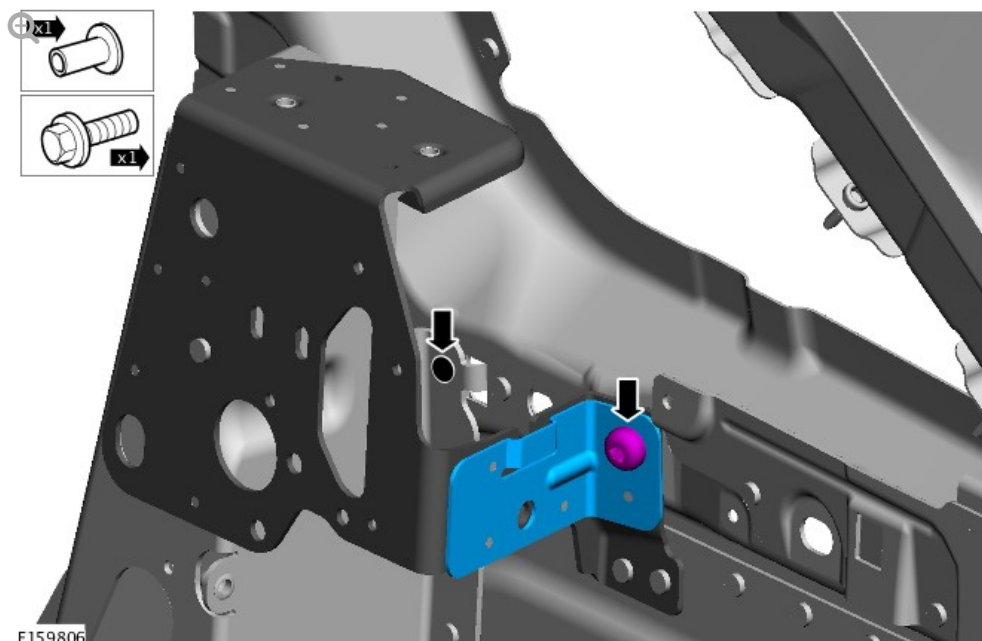
Remove the SPR's as indicated.

24.



Remove the SPR's as indicated.

25.

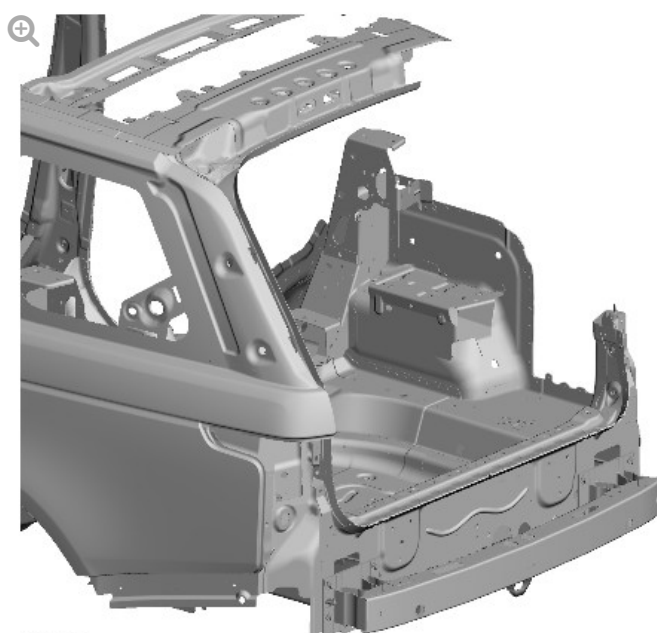


Remove the SPR and the bolt as indicated.

26.

CAUTION:

Make sure that when the old inner quarter panel upper is removed the remaining panels are supported avoiding any damage until the new inner quarter panel upper has been installed.



Separate the panels.

INSTALLATION

CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
- SPR installation requires using the Land Rover approved SPR tool.

1. Remove rivet remnants.

2. Dress flanges where necessary.

3. Using a fine bristle disc, clean and prepare the panel surfaces.

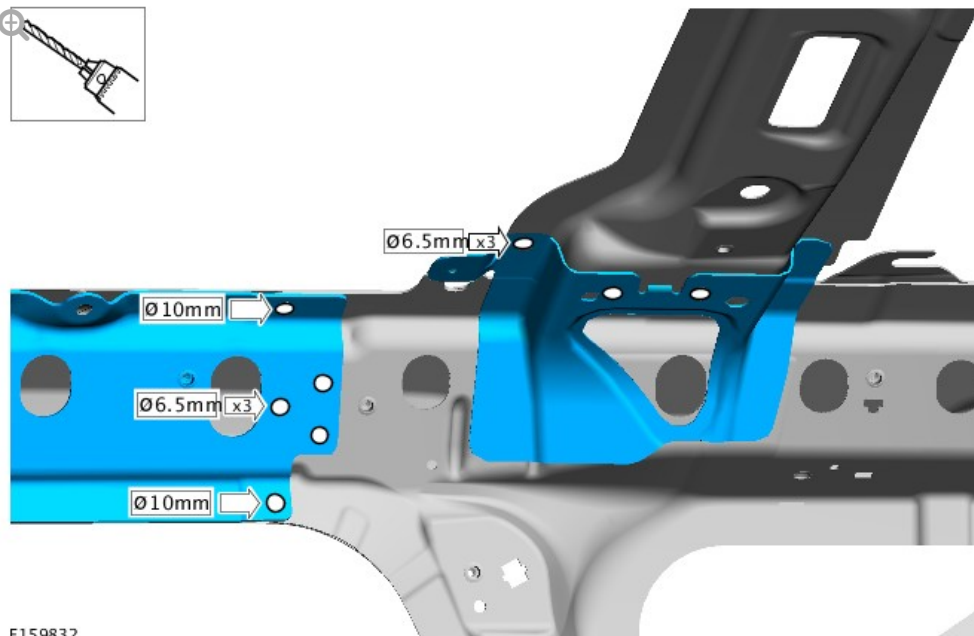
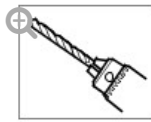
4.

CAUTION:

This step requires the aid of another technician as the panel will be heavy.

Offer up the new inner quarter panel upper and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.

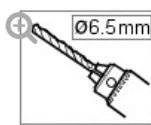
5.



E159832

Drill holes where the BSF's and the MIG plug welds are to be installed as indicated.

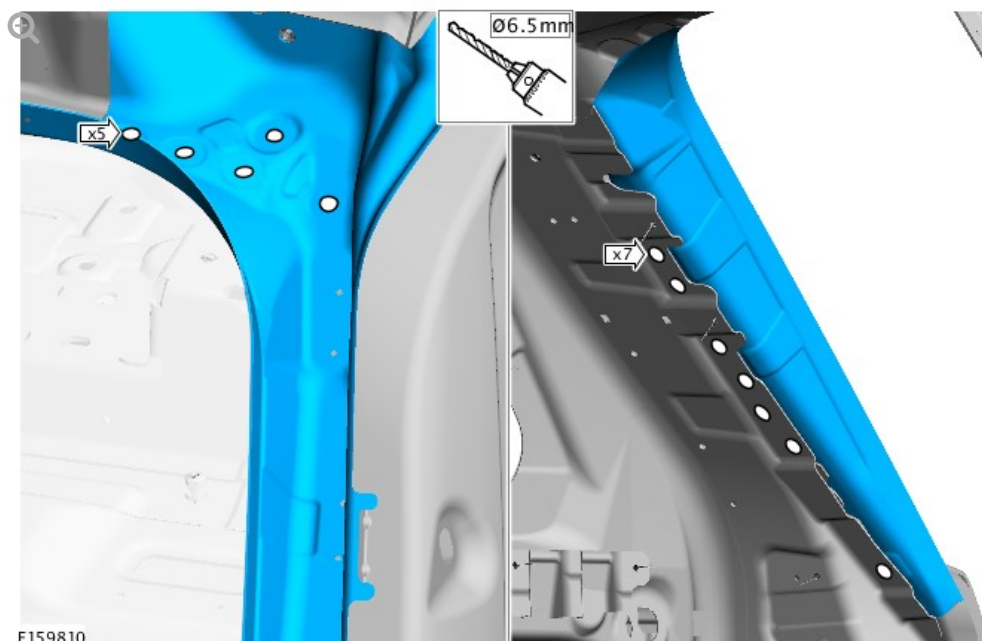
6.



E 150986

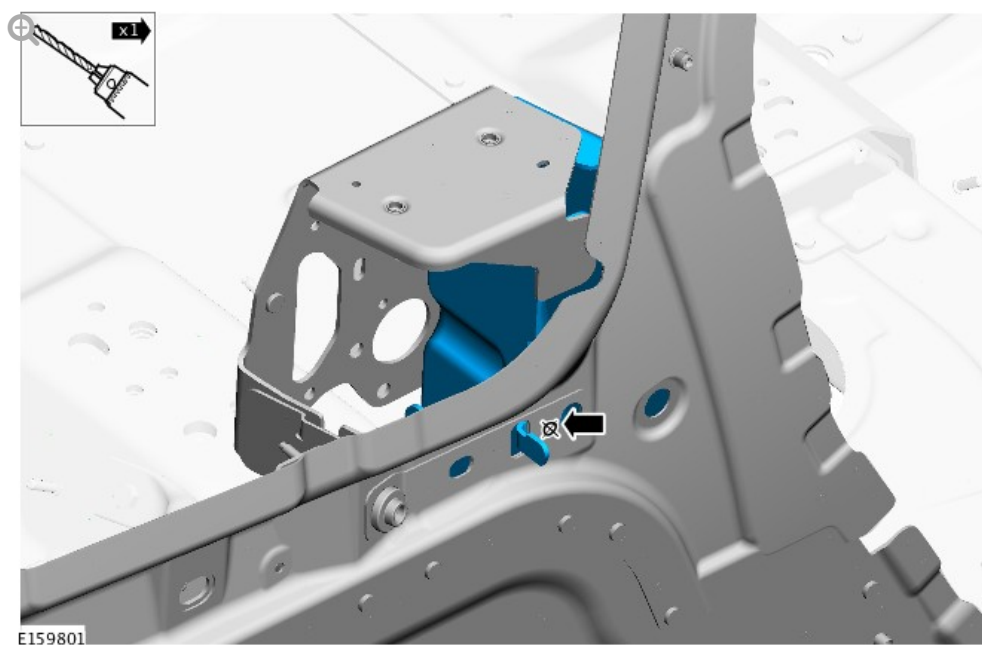
Drill holes where the BSF's are to be installed as indicated.

7.



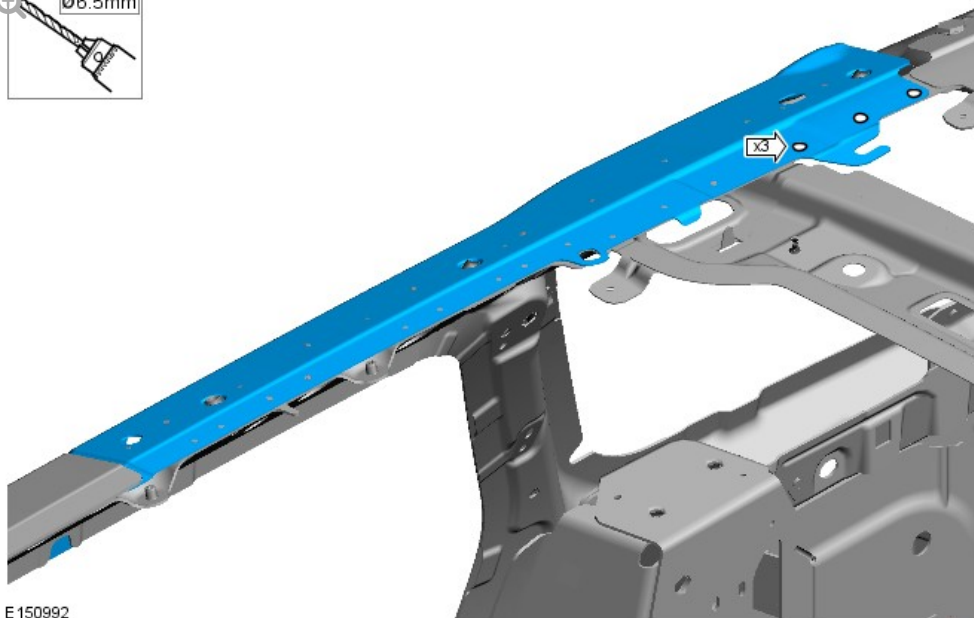
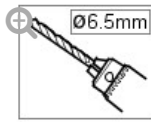
Drill holes where the BSF's are to be installed as indicated.

8.



Drill holes where the BSF is to be installed as indicated.

9.



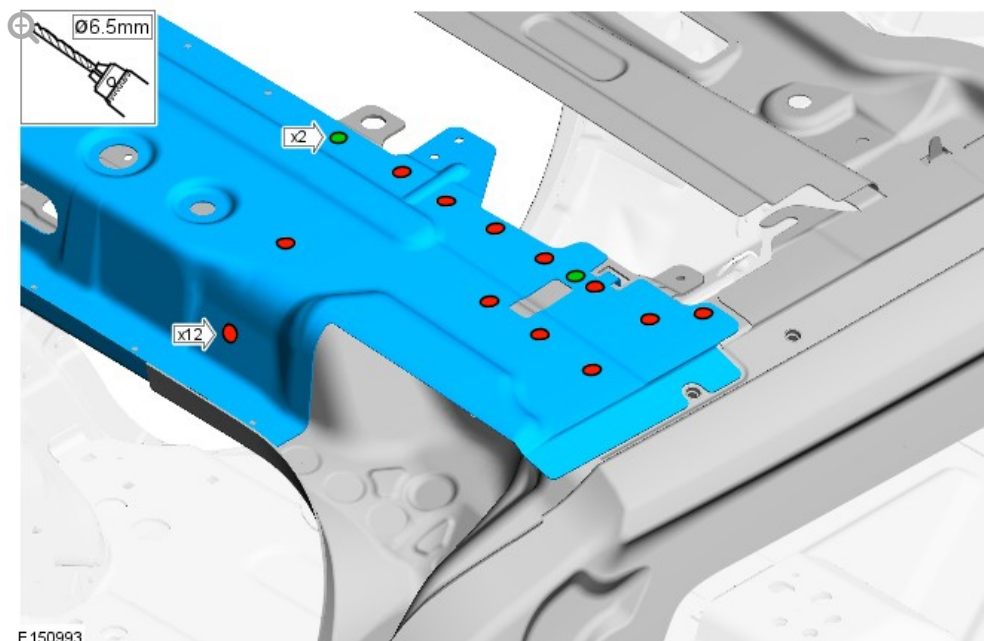
E 150992

Drill holes where the BSF's are to be installed as indicated.

10.

NOTES:

- The **green holes** indicate the holes are drilled from the **inside out**.
- The **red holes** indicate the holes are drilled from the **outside in**.



Drill holes where the BSF's are to be installed as indicated.

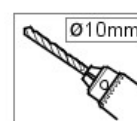
11.

CAUTION:

This step requires the aid of another technician as the panel will be heavy.

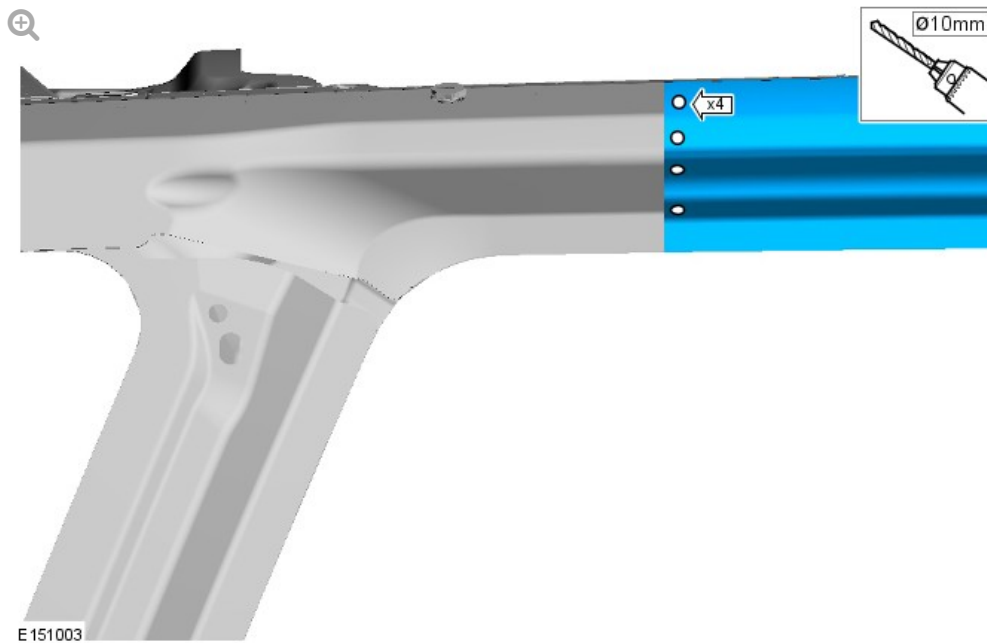
Remove the new inner quarter panel upper from the vehicle.

12.



Using a 10mm drill bit, drill holes for the metal inert gas (MIG) plug welds ready for the installation of the backing plate as indicated.

13.



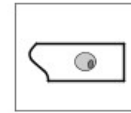
Using a 10mm drill bit, drill holes for the MIG plug welds ready for the installation of the backing plate as indicated.

14. Deburr the drilled holes.

15. Measure and cut two 50mm wide backing plates, 25mm each side of the MIG butt joint, from the quarter panel remnant.

16. Cut four run on/run off tabs from the quarter panel remnant.

17.



MIG plug weld the backing plate to the quarter panel as indicated.

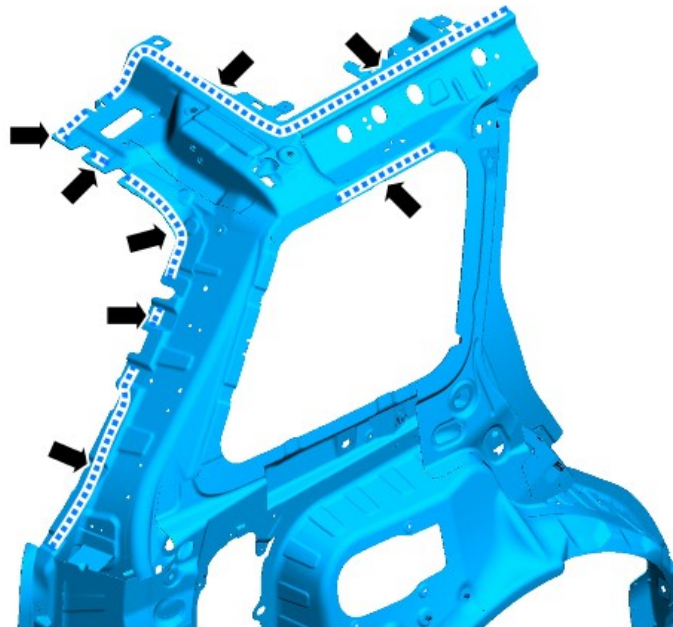
18. Using a fine bristle disc, clean and prepare the panel surfaces.

19. Apply the coupling agent where the Land Rover recommended bonding material is to be applied and allow to dry.

20.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.



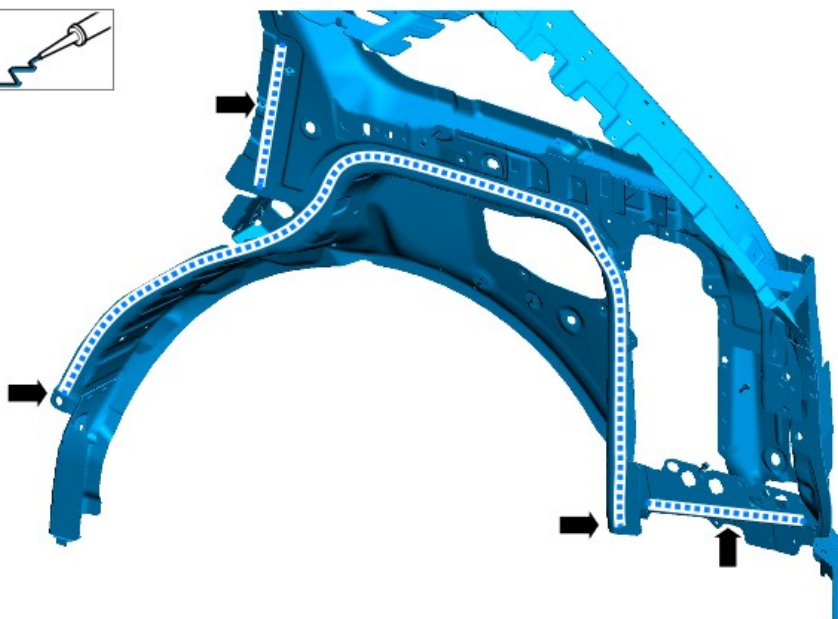
E 150994

Apply a 5mm zig zag bead of Land Rover recommended bonding material to the body joints as indicated.

21.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.



E 150995

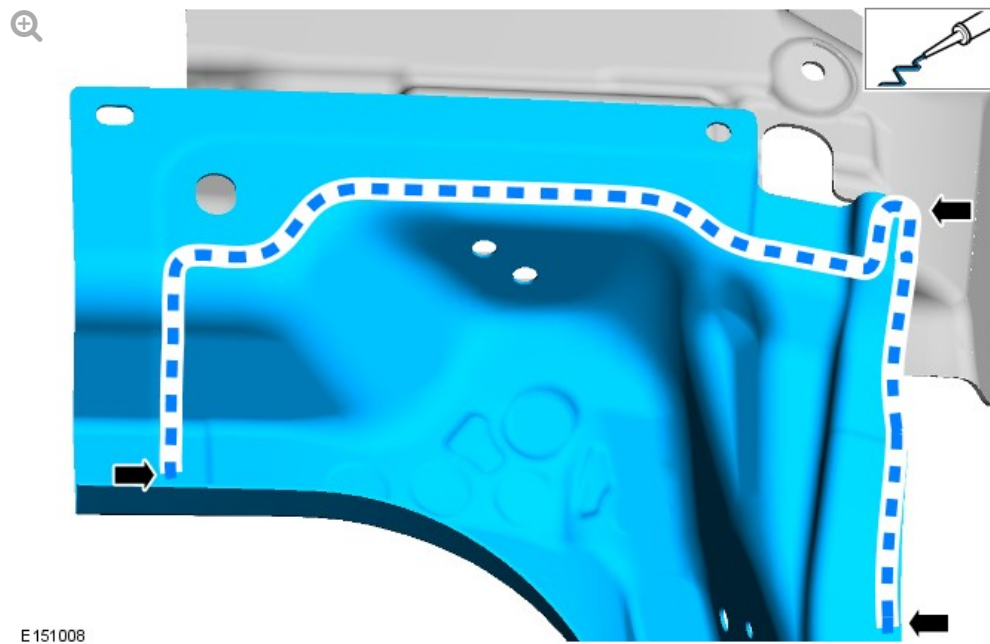
Apply a 5mm zig zag bead of Land Rover recommended bonding

material to the body joints as indicated.

22.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.



Apply a 5mm bead of Land Rover recommended bonding material to the body joints as indicated.

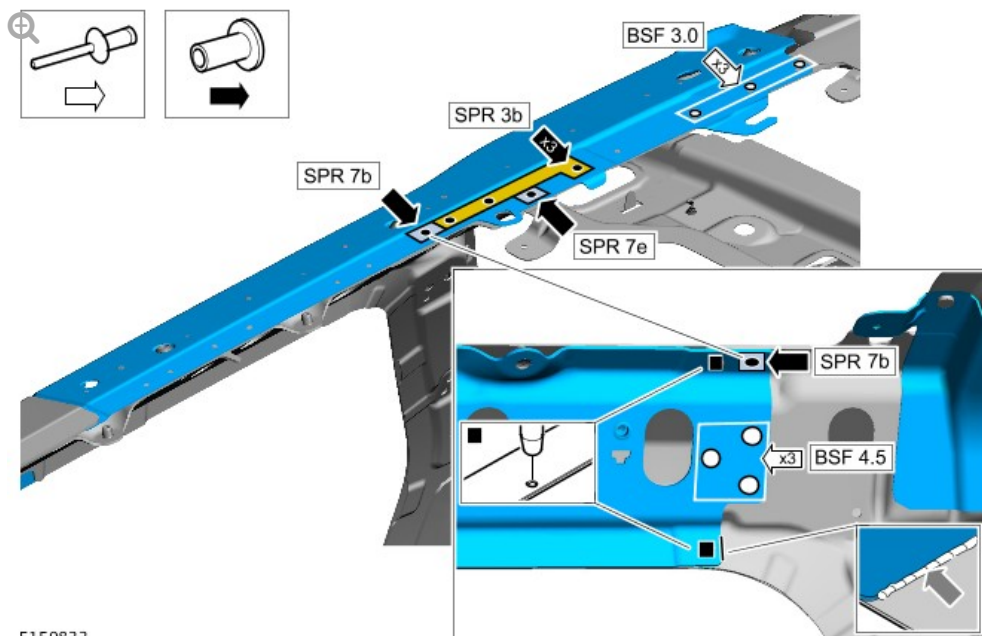
23.

CAUTION:

This step requires the aid of another technician as the panel will be heavy.

Offer up the new inner quarter panel upper and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.

24.



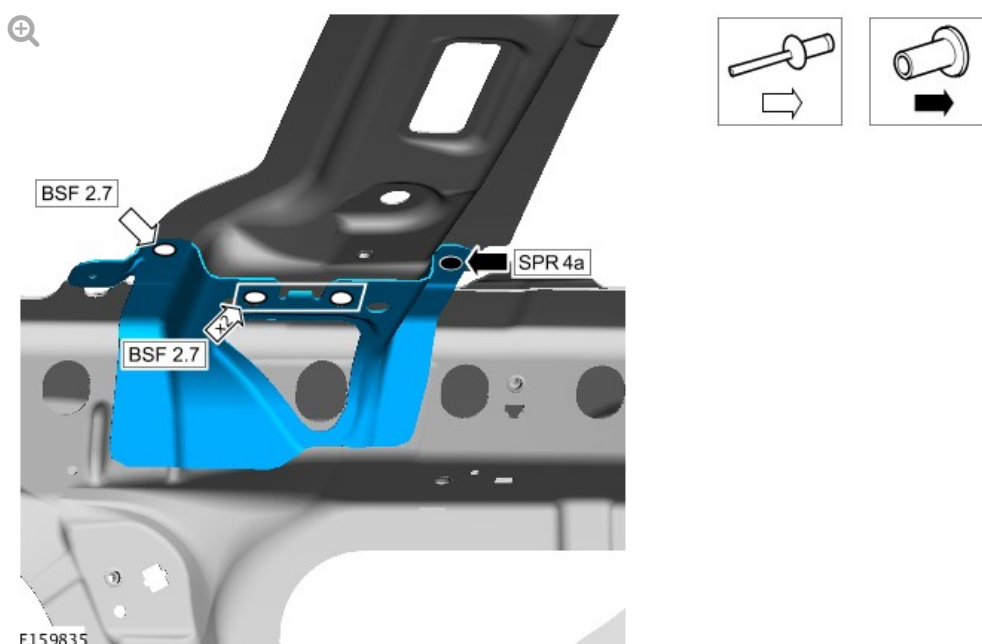
E159833

Install the SPR's and the BSF's as indicated. MIG plug and seam weld as indicated.

25.

NOTE:

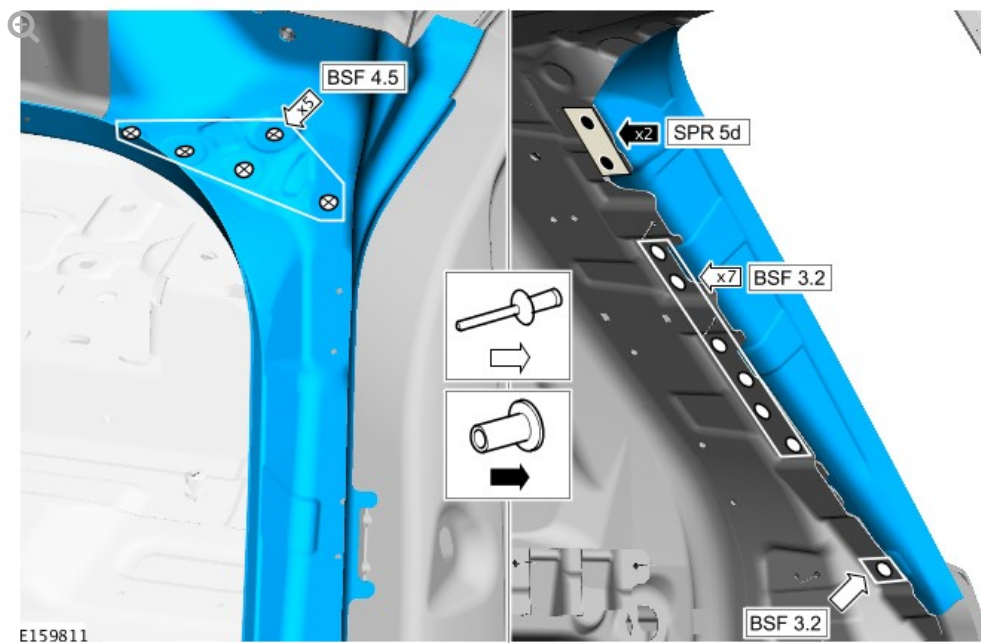
Make sure that the SPR is installed in conjunction with the roof rear panel.



E159835

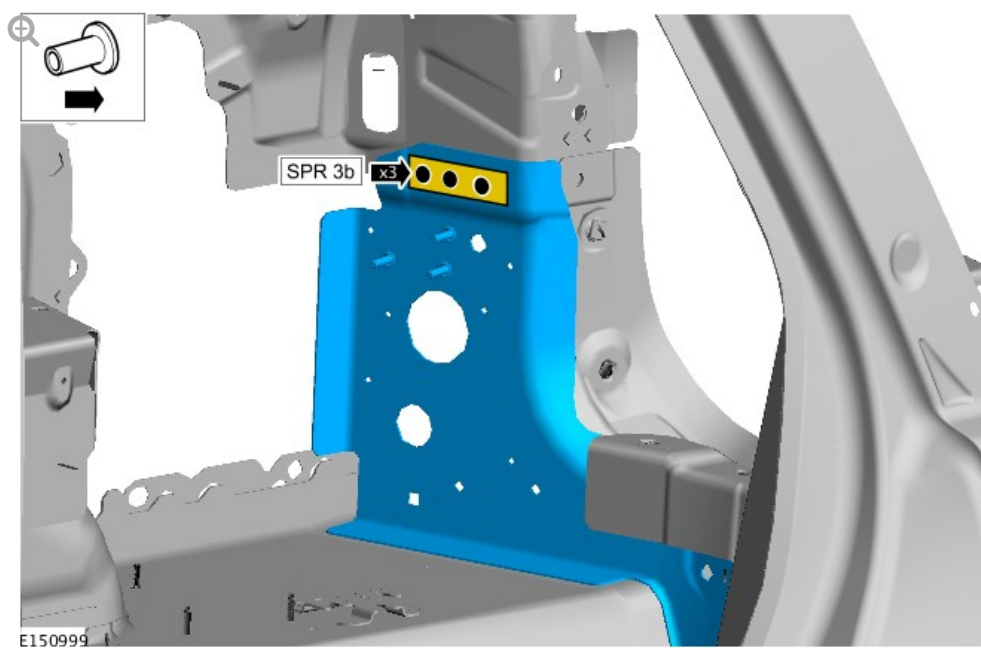
Install the BSF's as indicated.

26.



Install the SPR's and the BSF's as indicated.

27.

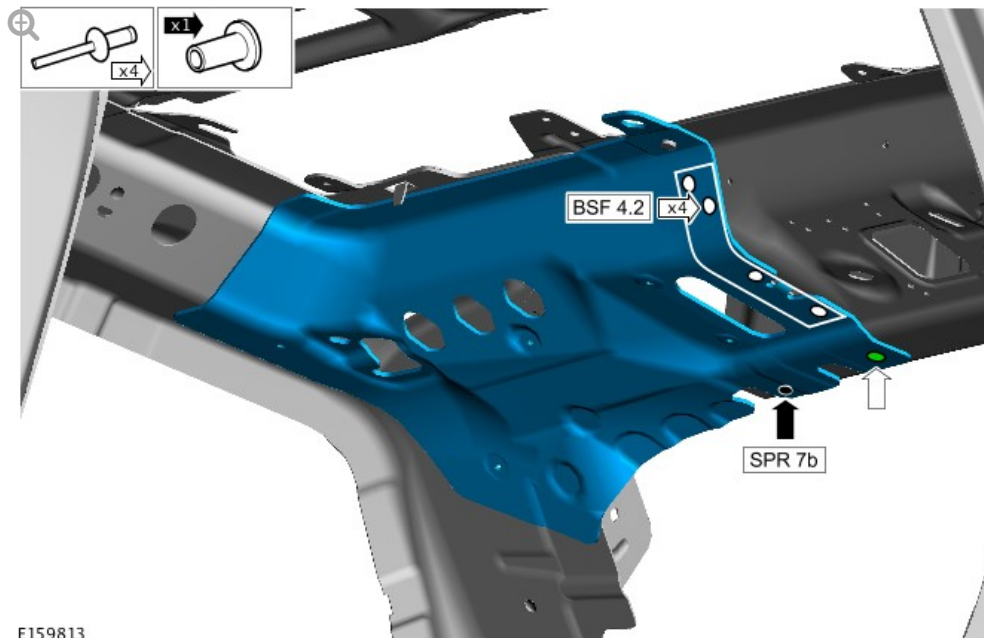


Install the SPR's as indicated.

28.

NOTE:

Make sure that the BSF indicated by the **green** hole is **not installed** as this carried out in the roof installation.

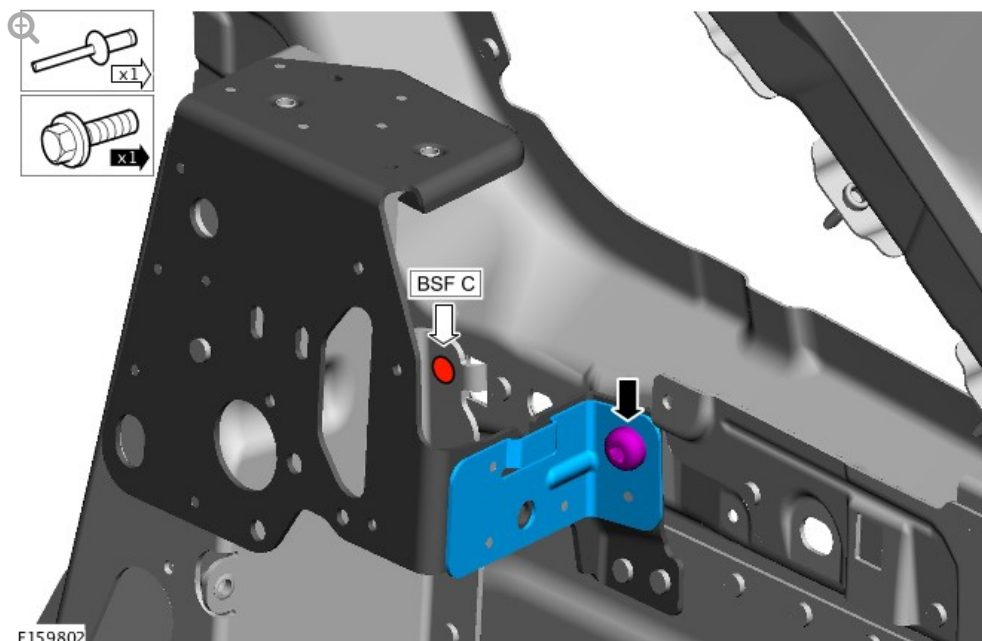


Install the SPR's and the BSF's as indicated.

29.

NOTES:

- Make sure that the BSF highlighted in **red** is installed from the **outside** in.
- Make sure that the **bolt** is **renewed**.



Install the BSF and the bolt and tighten to the new bolt to 40Nm as indicated.

30. Install the quarter panel.

31. Tack weld the run-on/run-off tabs to the installed quarter panel.



MIG plug weld the backing plates and MIG weld the quarter panel section butt joint as indicated.

33.

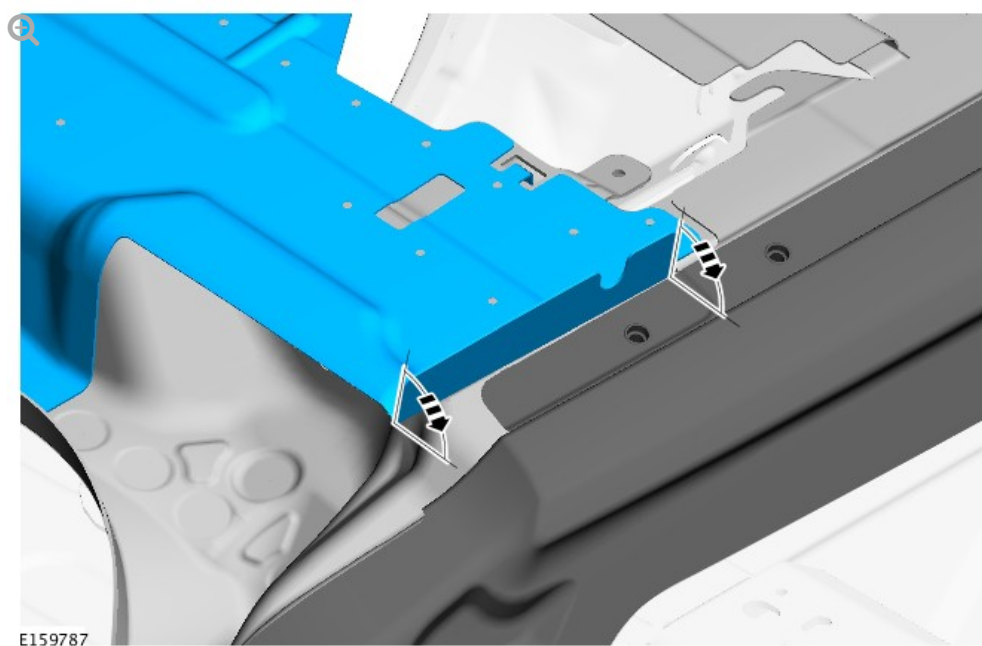
NOTE:

Make sure that the rivets highlighted with a cross are not installed at this stage as they are installed in conjunction with the roof panel.

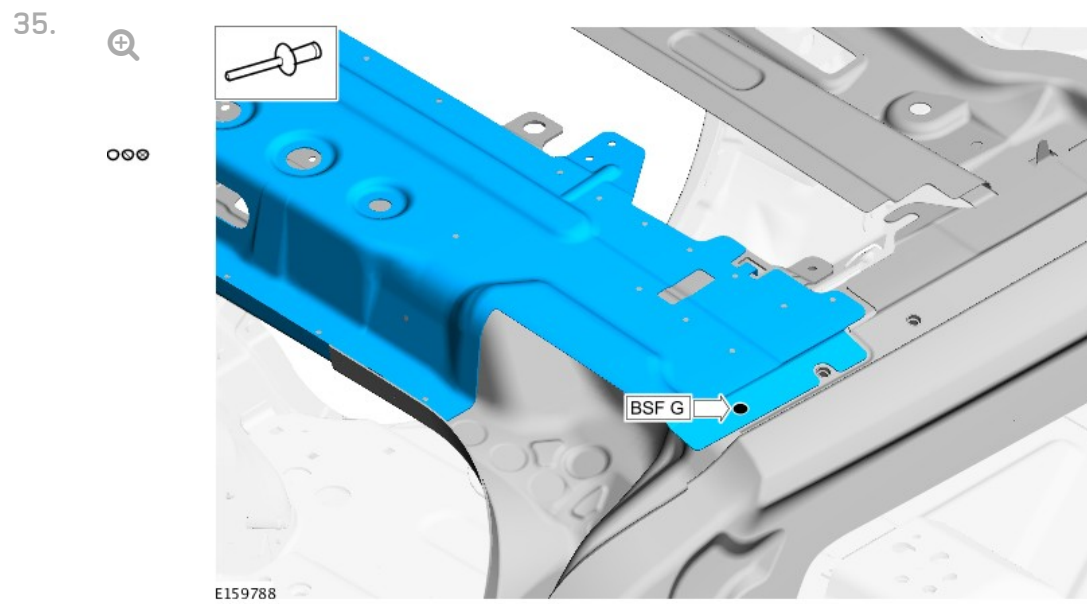


Install the BSF's and SPR's as indicated.

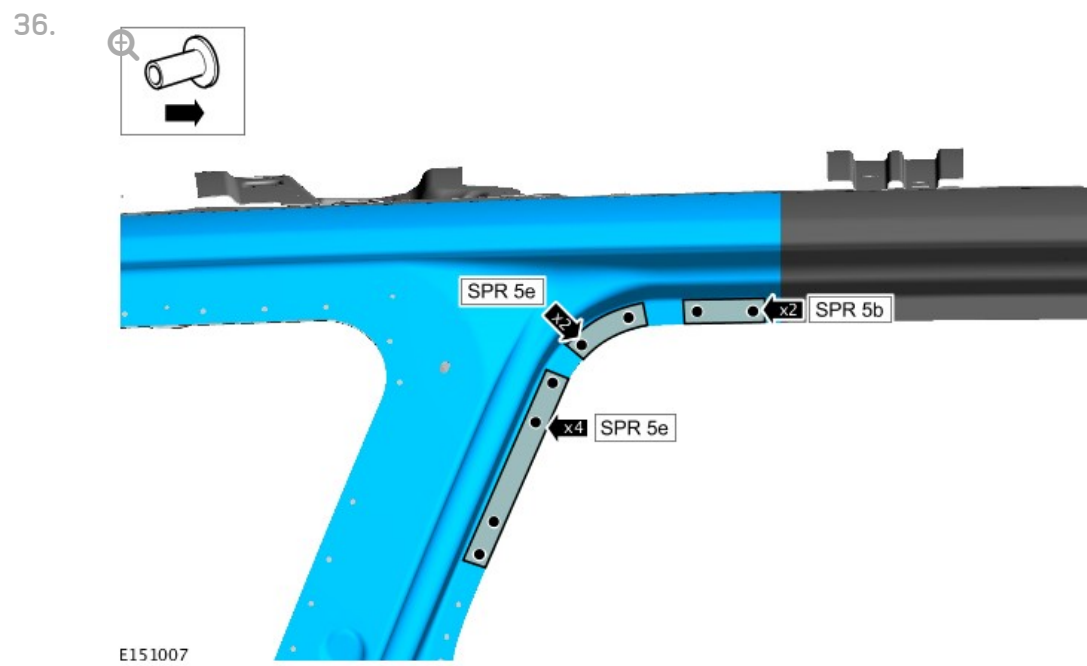
34.



Fold down the edge of the rear roof header rail as indicated.

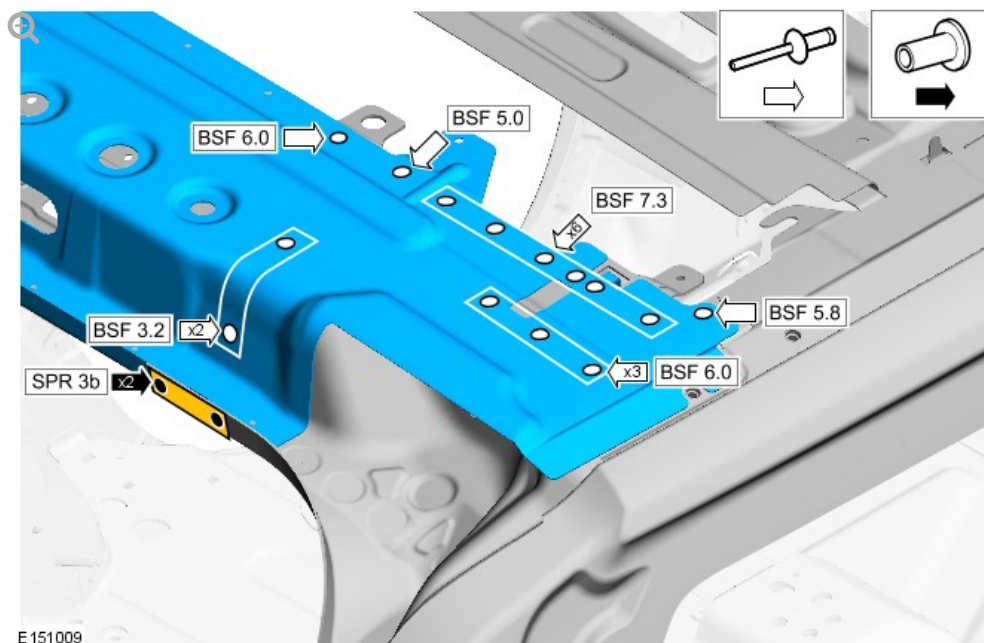


Install the BSF as indicated.



Install the SPR's as indicated.

37.



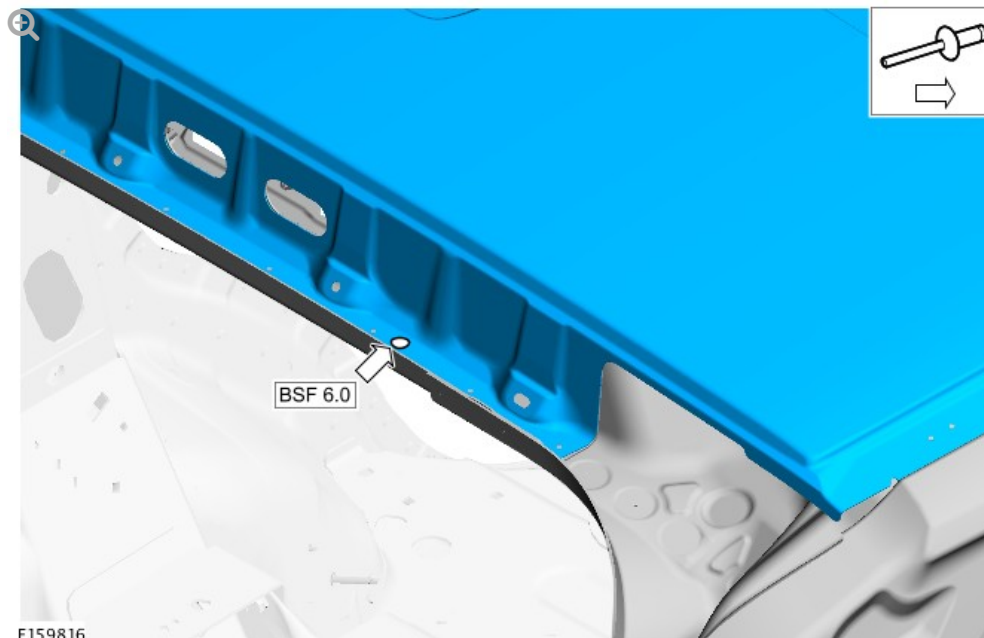
Install the BSF's and the SPR's as indicated.

38. Install the roof panel.

39.

NOTE:

Make sure that the BSF is installed after the installation of the roof panel and from the inside out.



Install the BSF as indicated.

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40. Remove any excess adhesive.
-
41. Make sure that any open or exposed panel joints are suitably sealed.
-
42. Make sure corrosion protection is applied to all areas affected by repair.
-
43. The installation of associated panels and components is the reversal of removal procedure.